

WASP-12b

R-0555

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_171225 · created 2026-07-23T13:45:13+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458112.84555 +/- 0.00046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.132 +/- 0.016
Transit depth (Rp/Rs)^2	1.75 +/- 0.41 [%]
Orbital Inclination (inc)	86.72 +/- 1.85
Airmass coefficient 1 (a1)	1.002 +/- 0.011
Airmass coefficient 2 (a2)	0.001 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	#3 - [467, 36]
Optimal Aperture	4.55
Optimal Annulus	4.93
Transit Duration (day)	0.1307 +/- 0.0027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.132 ± 0.016 vs 0.1178 (deviation 0.89σ)
Duration fit vs published	0.1307 d vs 0.125 d (deviation 2.11σ)
Mid-time O–C	0.8 min
Depth SNR	8.2
Residual scatter	1.09 %

Light curve

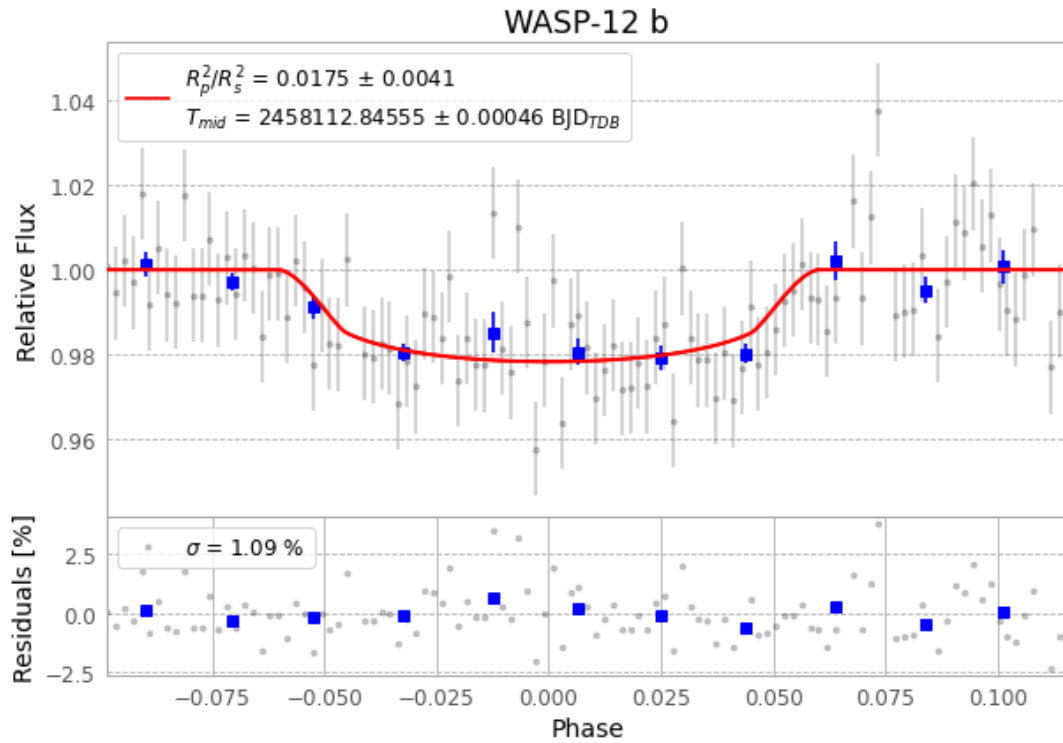


Figure 1 — FinalLightCurve_WASP-12 b_2017-12-24.png (SHA-256 cf2d4f68903eb9ea...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 264], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 74a5e56cfd897748...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

113 FITS frames (74 MB), WASP-12171225053312.FITS → WASP-12171225110912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-171225.log (b1a8c4f89ae4...), FinalLightCurve_WASP-12 b_2017-12-24.png (cf2d4f68903e...), FinalLightCurve_WASP-12 b_2017-12-24.csv (bc3c5667b239...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-23 13:45Z.